11
The emergence of prepositional genitives in Greek and its diachronic implications

Dimitris Michelioudakis, Stergios Chatzikyriakidis, and Giorgos Spathas

11.1 Introduction

The ongoing gradual loss of inflectional genitives and the concomitant rise of prepositional genitives in several Northern Greek varieties gives rise to several theoretical and diachronic questions regarding the interaction of syntax and morphology. A prominent issue is whether in such cases the adposition takes on all the syntactic properties and functions of the genitive inflection and whether this change means the wholesale replacement of one syntactic mechanism by another. More importantly, seen from a broader historical and comparative perspective, the study of this development bears on the conditions of retaining or losing the inflectional marking of case and on the conditions of developing an adpositional realization for non-core cases.

So far, little is known about the preconditions of either the retention or the loss of genitive inflection. The descendants of languages with inflectional genitives (e.g. Ancient Greek, Latin, Proto-Slavic, Proto-Germanic) display no uniformity with respect to the current realization and distribution of genitive arguments. This is evident if one compares, for instance, Western Romance to Modern Greek or to most of the contemporary Slavic languages. The amount of morphological simplifications, syncretisms, and losses in the history of each language does not seem to be a safe predictor. The loss of nominative-accusative distinctions in all nominal morphology is indeed something that all Western Romance languages have in common, alongside the loss of the inflectional genitive and the development of a prepositional genitive. This property strongly differentiates Western Romance from e.g. Russian, Polish, Serbo-Croatian, and Standard Greek, which retain the inflectional genitive. Nevertheless, this lack of morphological nominative-accusative contrasts cannot account for the productive use of inflectional genitives in languages such as Romanian (cf. nom/acc: *o casă* 'a house', gen/dat: *unei case* 'of/to a house'). Likewise, the Northern Greek varieties currently losing their inflectional genitive are likely to end up distinguishing the nominative from the accusative on nouns, while lacking a morphologically distinct

Dimitris Michelioudakis et al., *The emergence of prepositional genitives in Greek and its diachronic implications.* In: *The place of Case in Grammar.* Edited by: Christina Sevdali, Dionysios Mertyris, and Elena Anagnostopoulou, Oxford University Press. © Dimitris Michelioudakis et al. (2024). DOI: 10.1093/oso/9780198865926.003.0011

OUP UNCORRECTED PROOF – FIRSTPROOFS, Wed. Feb 14 2024, INTEGRA

genitive. Contrasting varieties of Greek that differ minimally, one of the differences being the productivity of the genitive, we can shed more light on the morphological and syntactic correlates of this kind of change.

In this chapter, we establish that a number of Greek varieties exhibit a productive prepositional syntax of genitive arguments, which has no precedent in the history of the language. This pattern extends well beyond suppletion, i.e. the prepositional variant cannot be argued to be a last resort realization of genitive arguments when the respective noun lacks inflectional genitive forms. Specifically, such genitive PPs are headed by *apo* 'from'. This is the preposition that Standard Modern Greek (henceforth SMG) employs to express sources and partitivity as well as *by*-phrases, but not possessors or non-agent arguments of event nominals. Instead, the Northern Greek varieties in question (henceforth NG) can realize all sorts of adnominal argument as PPs consisting of *apo*+DP_{acc} (1).

(1) i fotoɣrafia apo ta peðja
the picture from the kids.ACC

'the picture (a) coming from/sent by the kids (SMG/NG); (b) belonging to the kids (NG/*SMG); (c) depicting the kids (NG/*SMG)'

We further establish that such a full-blown prepositional frame instantiates a radically different syntactic mechanism and does not merely replicate the distribution of the lost genitive. We suggest that the emergence of a non-affixal morpheme realizing the genitive (and possibly non-core cases in general) is only made possible in the history of a language once Case and Number are no longer fused. When the genitive inflection itself carries cues of Case-Number fusion, its loss is arguably favoured or necessitated by the overall reduction of other such cues and this in turn facilitates the rise and generalization of the adpositional variant.

The chapter is organized as follows: in Section 11.1 we present the syntax of genitive DPs and clitics in SMG, in Section 11.2 the Northern Greek patterns are presented and contrasted to the SMG data, while in Section 11.3 we identify aspects of nominal declension the observed variability correlates with and offer a morphosyntactic analysis with diachronic implications. In Section 11.4 we summarize our conclusions.

11.2 The SMG pattern

In SMG, nouns, adjectives, determiners, and pronouns have a morphologically distinct genitive. The unmarked position of genitive-marked DPs is postnominal (2). Focus-fronting or topic-fronting places genitive DPs to the left of the head noun and its determiner, if any (3), see also Alexiadou et al. (2007: 580–3) and references therein.

OUP UNCORRECTED PROOF – FIRSTPROOFS, Wed. Feb 14 2024, INTEGRA

(2) a. to (oreo) spiti tu ðaskalu
the nice house the.M.GEN teacher.GEN

b. *to (oreo) tu ðaskalu spiti
the nice the.M.GEN teacher.GEN house

'the teacher's (nice) house'

(3) {tu ðaskalu / tu ðASKALU} to spiti
the.M.GEN teacher.GEN the.M.GEN teacher.GEN the house

'the teacher's house'

Typically, there can only be one genitive DP per head noun. For instance, complex event nominalizations (see Grimshaw 1990) of two-place predicates of the sort exemplified in (4–5) only allow, or indeed require (see Alexiadou 2001: 78ff), the internal argument to be realized as a genitive DP.[1] The external argument can be

[1] An additional genitive DP is apparently allowed in some result nominals, as long as this additional DP has a possessor or author interpretation (i). In such cases, the first genitive (the one that serves as a 'complement', using Grimshaw's (1990) terminology) does not display quite the same behaviour as adnominal arguments. For instance, no modifiers can intervene between the head noun and the first genitive (ii). In contrast, modifiers of different sorts (PPs, AdjPs etc.) can easily intervene between the head noun and an argumental genitive DP in a complex event nominalization (iii). Also, the same modifiers that are allowed between a result nominal and its possessor/author are also allowed to appear between the first genitive and the possessor/author genitive (iv).

(i) i / mia metafrasi tis Oðisias tu Kakriði
the/ a translation the.GEN Odyssey.GEN the.GEN Kakridis.GEN

'the/a translation of the Odyssey by Kakridis' (from Alexiadou et al. 2007: 589)

(ii) *mia metafrasi kali tis Oðisias tu Kakriði
a translation good the.GEN Odyssey.GEN the.GEN Kakridis.GEN

'a good translation of the Odyssey by Kakridis'
(only interpretation allowed: 'A good translation of Kakridis's Odyssey')

(iii) i anakrisi (epi pende ores / i prosektiki) tu martira
the interrogation for five hours the careful the.GEN witness.GEN

'the careful interrogation of the witness/the interrogation of the witness for five hours'

(iv) a. ena vivlio {kenurjo / ja efivus} tu Kakriði
a book new/ for teenagers the.GEN Kakridis.GEN

'A new book by Kakridis/a book for teenagers by Kakridis'

b. mia metafrasi tis Oðisias {kenurja / ja efivus} tu Kakriði
a translation the.GEN Odyssey.GEN new for teenagers the.GEN Kakridis.GEZ

These observations suggest that, of the two genitives in (i) and (ivb), only the outmost one occupies the canonical unique position in which genitive Case is licensed. Examples (iii) and (iva) either suggest that this position can be preceded by certain postnominal modifiers, or that right dislocation is possible from the canonical position for genitive DPs. Even in the latter case, this dislocation operation cannot target the innermost genitive (ii). This kind of dislocation can preserve a restrictive interpretation, therefore it does not involve loose apposition of a second DP with an empty head noun. The innermost genitive seems to have a special status, licensed under different conditions, under strict adjacency with the head noun. Kolliakou (1999) calls it a pseudo-possessive, thus contrasting it to true possessives, which are unique in all types of DP, while Markantonatou (1992) argues that the head noun and the innermost genitive form some sort of compound. Indeed, the rigidity of this unit points to it being X^0, rather than more complex.

OUP UNCORRECTED PROOF – FIRSTPROOFS, Wed. Feb 14 2024, INTEGRA

left unexpressed (4) or be realized as a *by*-phrase, headed by the preposition *apo* 'from' (5).

(4) i (jenea/polimini) ðierevnisi (*tu Jani) tu atiximatos
the brave/lengthy investigation the.GEN John the.GEN accident.GEN
(*tu Jani)
the.GEN John
'John's (courageous) investigation of the accident (for many months)'

(5) i (jenea/polimini) ðierevnisi tu atiximatos apo
the brave/lengthy investigation the.GEN accident.GEN from
ton Jani
the.ACC John
'the (courageous) investigation of the accident by John for many months'.

A further option regarding the realization of the external argument in such nominals is its realization as a pronominal clitic, typically adjacent to a prenominal adjective (6).

(6) i jenea tu polimini ðierevnisi tu
the courageous his.M.GEN.CLITIC lengthy investigation the.GEN
atiximatos
accident.GEN
'his courageous investigation of the accident for many months'

In fact, genitive pronominal clitics can be used for possessors and adnominal arguments of all sorts. When a nominal has a unique overt possessor or adnominal argument, this can be realized as a pronominal clitic which appears either postnominally, adjacent to the noun, or prenominally, adjacent to a prenominal adjective (7a–b). In complex event nominalizations of two-place predicates, a genitive pronominal clitic can co-occur with a genitive DP, as already seen in (6), or (slightly more marginally) with another genitive clitic (8).

(7) a. i proti eksetasi {tu / tu Jani}
the first examination his.GEN.CLITIC / the.M.GEN John.GEN

b. i proti {tu / *tu Jani} eksetasi
the first his.GEN.CLITIC the.M.GEN John.GEN examination
'John's/his first examination' (John/he = agent, patient, or possessor)

(8) a. i proti mu poliori eksetasi tis
the first my.GEN.CLITIC lengthy examination the.FEM.GEN
asthenus
patient.GEN
'my first examination of the patient that lasted many hours'

OUP UNCORRECTED PROOF – FIRSTPROOFS, Wed. Feb 14 2024, INTEGRA

b. i proti mu poliori eksetasi tis
the first my.GEN.CLITIC lengthy examination her.GEN.CLITIC
'my first examination of her that lasted many hours',

When a clitic co-occurs with a genitive DP, two restrictions must be obeyed: (a) the clitic must precede the postnominal genitive DP and is typically prenominal, as in (6); (b) the genitive DP must realize the lower argument and the clitic must realize the higher argument and not vice versa. When two clitics co-occur, one is necessarily prenominal and one is necessarily postnominal. Similarly to the co-occurrence with genitive DPs, the prenominal clitic must correspond to the higher argument and the postnominal clitic can only correspond to the lower argument. In other words, up to two adnominal genitive arguments can co-occur, and at least one of them must be realized as a clitic. It can thus be argued that postnominal clitics occupy the same structural position as genitive DPs.

As already implied, the possible broad semantic roles that can be borne by adnominal genitive DPs are: internal argument/theme, external argument (agent or author) and possessor, i.e. an individual bearing a pragmatically inferrable relationship to the referent of the head noun other than the ones mentioned previously, e.g. ownership. There are two types of nominal constructions that are compatible with any of the three roles: picture nominals (with a genitive potentially realizing the possessor, the author or the depicted individual) and nominals ambiguous between a process and a result reading. As can be seen in (9), when two adnominal genitive arguments co-occur, the way they are ordered strictly respects the thematic hierarchy POSSESSOR > AGENT/AUTHOR > THEME (Cinque 1980; Pollock 1989a).

(9) i proti mu periγrafi/fotoγrafia tu Man Rei
the first my.GEN.CLITIC description/picture the.M.GEN Man Ray
'the first description/picture I made/took of Man Ray'
(AUTHOR/AGENT>THEME)
'the first description/picture I came to own of Man Ray'
(POSSESSOR>THEME)
'the first description/picture I came to own by Man Ray'
(POSSESSOR>AUTHOR)
'*the first description/picture of me that Man Ray made/took'
(*THEME>AUTHOR)
'*the first description/picture of me that Man Ray bought'
(*THEME>POSSESSOR)
'*the first description/picture by me that Man Ray bought'
(*AUTHOR>POSSESSOR)

The same ordering effects also obtain if we replace the genitive DP with a genitive clitic.

OUP UNCORRECTED PROOF – FIRSTPROOFS, Wed. Feb 14 2024, INTEGRA

These effects suggest that the two positions occupied by the co-occurring arguments are not the positions where they are assigned their thematic role, as neither position is uniformly mapped onto one role. We should rather assume that their unmarked surface positions are two functional positions attracting adnominal arguments from the thematic domain in an order-preserving fashion, i.e. obeying Superiority (Chomsky 1973). We will therefore be assuming that there is a low area in which adnominal arguments are first merged and a higher area into which up to two arguments can be attracted. The higher area in Greek contains functional projections hosting prenominal adjectives (see Cinque 2010), a functional projection that attracts the head noun (via head-movement of *n* or *n*P-movement) and, crucially, two functional projections for adnominal genitives: (i) a projection $F_{GEN}P$ immediately below the surface position of the noun, which hosts genitive DPs and clitics, and (ii) a projection $F_{CLITIC}P$, above the projection hosting the noun, which only hosts clitics (10). This latter position must be immediately preceded by an adjective, which provides the phonological host onto which the (en)clitic pronoun cliticizes. The adjective immediately preceding the clitic is typically the highest available in the DP (e.g. the evaluative adjective in (6) or the ordinal adjective in (8) and (9)), or the dummy adjective *ðikos/-i-/o* ('own'), which yields emphatic or contrastive pronominal possessors, i.e. *ðikos/-i-/o mu* 'my own', *ðikos/-i/-o su* 'your own' etc. (see Alexiadou 2005 for a synchronic and diachronic account of this high position, based on Manolessou 2000, alongside an account of the *ðikos/-i/-o* construction). It is then reasonable to assume that $F_{CLITIC}P$ is a high projection, above all projections ~~which~~ host AdjPs~~, which~~ triggers either (a) attraction of the highest adjective, if there is one~~, to its left~~ at PF or (b) insertion of the dummy adjective *ðikos/-i/-o* (see 13). We will also assume that $F_{GEN}P$ and $F_{CLITIC}P$ are not obligatorily part of every DP and that it is possible to build a DP containing/activating either one of them, both of them, or neither of them.

(10) $[_{DP}$ D (Adj) [$F^0{}_{CLITIC}$ (Adj) n [DP_{GEN} $F^0{}_{GEN}$ [ext. arg. <n> v int.arg]]]]

Regarding the status of the various adnominal arguments/complements, we follow here Grimshaw's (1990) tripartite distinction of nominals into (a) complex event/argument supporting nominals, (b) simple event nominals, and (c) result/R-nominals. Nominals of type (b) and (c) are syntactically similar, in that genitive DPs are optionally realized, while in nominals of type (a), which can be diagnosed by e.g. the availability of aspectual modification, the internal argument is obligatorily realized.

In the case of complex event nominals in particular, the aforementioned lower area may be considered to be a verbal constituent embedded under *n* (see Alexiadou 2009, Alexiadou et al. 2007 among others), although we need not make any specific assumptions here about the category of the argument introducing projections. In particular, what we call here the external argument is arguably introduced

OUP UNCORRECTED PROOF – FIRSTPROOFS, Wed. Feb 14 2024, INTEGRA

in the same position as possessors of result nominals, rather than the specifier of a v/Voice head. This is Alexiadou's (2017b) conclusion, which is based on Portner's (1992) observation that subjects of two-place predicate nominalizations never scope below negation, unlike transitive clausal subjects, which originate VoiceP-internally. We will further assume that possessors too originate lower than their surface position, specifically in the specifier of *n*, with *n* originating below F^0_{GEN}.

Then, the unmarked order 'head noun > genitive' is the result of two movement operations. First, a genitive DP originating in any of the aforementioned positions (vP internal projections or spec-*n*) moves into Spec-F^0_{GEN}. Second, the surface placement of the head noun before the genitive is the result of successive head-movements from Root (to *v*) to *n* and then *n*- (or remnant nP-) movement just above F^0_{GEN}.

Below, we exemplify the analysis outlined previously. In (11), we have a unique postnominal genitive DP with a head noun which can be either a result nominal (i) or a complex event, i.e. argument supporting, nominal (ii). Only F^0_{GEN} is activated. In argument supporting nominals, the internal argument is obligatorily realized (cf. Grimshaw 1990). In order for it to be attracted from the lower thematic position, the external argument must be left implicit/unpronounced (and as such it causes no intervention effect; see Michelioudakis 2020, 2021 on implicit agents always bearing a subset of the probing features in F^0_{GEN} and thus causing no relativized minimality violation). In result nominals, there is no such restriction regarding the interpretation of the genitive DP. It can bear any pragmatically inferable relationship to the head noun, so one may postulate just one underspecified thematic position, possibly Spec-nP, from which genitives are attracted.[2]

(11)

	i	proti	periγrafi	tis	Kikis		
	the	first	description	the	Kiki.GEN		
i.	$[_{DP}$ D	(Adj) [	n	[DP_{GEN}		F^0_{GEN}	[poss./author <*n*> …]]]]
ii.	$[_{DP}$ D	(Adj) [	n	[DP_{GEN}		F^0_{GEN}	[ext. arg./poss. <*n*> *v* int.arg]]]]

i. 'the first description made/owned by/concerning Kiki'(result reading)
ii. 'the first description of Kiki'(only interpretation of the complex event reading)

As already said, F^0_{GEN} may also attract clitics (12).

[2] In a subclass of result nominals, namely picture nominals, there can also be an adnominal argument corresponding to the depicted entity/individual, alongside a possessor or an author. Possessors and authors can also co-occur as we saw in (9). All such arguments are ordered in accordance with Cinque's (1980) thematic hierarchy. So, in such cases, one must postulate positions for all three of them within the nP (possibly also within RootP), hierarchically ordered in the same way, even in the absence of the verbal structure that one can assume for complex event nominals.

OUP UNCORRECTED PROOF – FIRSTPROOFS, Wed. Feb 14 2024, INTEGRA

(12)

	i	proti	periγrafi	tis		
	the	first	description	her.GEN.CLITIC		
i.	[$_{DP}$ D	(Adj) [	n	[CLITIC$_{GEN}$	F$^0_{GEN}$	[poss./author <*n*> ...]]]]
ii.	[$_{DP}$ D	(Adj) [	n	[CLITIC$_{GEN}$	F$^0_{GEN}$	[ext. arg./poss. <*n*> *v* int.arg]]]]

i. 'the first description made/owned by her'
ii. 'the first description of her'

In (13), only F$^0_{CLITIC}$ is activated. Like F$^0_{GEN}$, it is able to attract a clitic from any thematic position, subject to the absence of any intervention effects.[3]

(13)

	i	proti	tis	periγrafi		
	the	first	her.GEN.CLITIC	description		
i.	[$_{DP}$ D	(Adj) [	F$^0_{CLITIC}$	n	[poss./author <*n*> ...]]]]	
ii.	[$_{DP}$ D	(Adj) [	F$^0_{CLITIC}$	n	[ext. arg./poss. <*n*> *v* int.arg]]]]	

i. 'the first description made/owned by her'
ii. 'the first description of her'

Finally, in (14) both positions are activated and attract two arguments in an order-preserving manner, so that the high clitic can never correspond to an argument thematically lower than the one realized by the postnominal genitive (DP or clitic).

(14)

	i	proti	mu	periγrafi	{ tis	Kikis	/ tis	}
	the	first	my.CLITIC	description	the	Kiki.GEN	her.CLITIC	
[$_{DP}$	D	(Adj)	[F$^0_{CLITIC}$	n	[DP$_{GEN\ /\ CLITIC}$		F$^0_{GEN}$	[ext. <*n*> *v* int.]]]]

'my first description of Kiki / '*Her first description of me'[4] 4

11.3 The Northern Greek pattern

In a number of varieties of Greek, including Standard Modern Greek, the plural genitive of certain nouns is falling out of use. As Mertyris (2014) notes, the two most prominent classes of such nouns are parisyllabic barytone feminine nouns in -a (e.g. *'bala* 'ball': **'balon* / **ba'lon* 'of the balls') and neuter diminutives (e.g.

[3] Following Michelioudakis 2020, 2021, non-generic implicit agents have the same set of *phi*-features as genitive clitics, thus giving rise to a relativized minimality violation and blocking attraction of the internal argument clitic by F$^0_{CLITIC}$. Indeed, constructions diagnosing implicit agents, e.g. control of the understood agent into temporal gerunds, are impossible in the presence of an internal argument genitive clitic and possible otherwise; i.e. when the internal argument is realized as a genitive clitic, the external argument is necessarily not projected, see Michelioudakis 2020, 2021 for details and examples.

[4] Another possible interpretation is 'the first (piece of paper/written text containing a) description by Kiki I came to own'. For such readings, we must postulate two distinct positions within nP, one for possessors and a lower one for authors of result nominals, see also fn. 2.

OUP UNCORRECTED PROOF – FIRSTPROOFS, Wed. Feb 14 2024, INTEGRA

peˈðaci 'little child': *peˈðacon/* peðaˈcon 'of the children'), with the latter being problematic in the singular too. Speakers of different varieties develop different avoidance strategies, one of which is the use of a PP headed by *apo* 'from'. In such varieties, prepositional genitives are in general in complementary distribution with inflectional genitives, serving as suppletive forms filling paradigmatic gaps, e.g. the ones mentioned or plural genitives in general (Kavakli/Eastern Rumelia and most Northern Greek varieties, Albanoudis 2014), or genitives of non-bisyllabic nouns (Roumloukiotika/Imathia, Deliopoulos 2008) and of unassimilated loan-words (Italiot/Calabria Greek, see Katsoyannou 1996, and more generally Mertyris 2014, 2019 and references therein for the dialectal distribution and diachronic development of such patterns).

More conservative varieties restrict the use of such prepositional substitutes to genitives which are also, more or less marginally, compatible with a partitive or source interpretation (15). Source and 'part-of' interpretations are the canonical interpretations of adnominal *apo*-PPs in SMG, alongside their use as *by*-phrases in complex event nominals (see (5)).

(15) a. Paratiro ta xerulja ap(o) tis tsajeres / ?*ton
I-watch the handles from the.ACC teapots.ACC the.GEN
tsa'jeron.
teapots.GEN
'I watch the handles of the teapots.'

b. To poði ap(o) to mosxaraki ine to kalitero kreas.
The leg from the.ACC calf/veal.ACC is the best meat
'The leg of the calf/veal is the best cut.'

Therefore, there are grammars that would accept (15a) and (15b) as felicitous genitive substitutes but would rule (16) out as odd, given the inferences that its source interpretation gives rise to.

(16) #Agikse to poði apo ton aðelfo mu.
Touched.3SG the leg from the brother.ACC my.clitic
'(S)he touched my brother's leg' (infelicitous unless the leg has been severed, i.e. literally coming/taken from my brother's body)

Nevertheless, (16) too is tolerated or deemed fully acceptable by many other speakers, especially from Northern Greece. In what follows we will focus on varieties in which this strategy is not subject to the restrictions that SMG poses and indeed varieties that extend this strategy well beyond cases of paradigmatic gaps. In such varieties, the prepositional genitive is not merely a suppletive/last resort form and is in fact available and preferred for all sorts of nouns, with the possible exception of proper names, and all the kinds of interpretation discussed in the previous section. More specifically, we will discuss data from a variety

OUP UNCORRECTED PROOF – FIRSTPROOFS, Wed. Feb 14 2024, INTEGRA

in which prepositional genitives are particularly productive, namely Grevena Greek (henceforth GG), spoken in the prefecture of Grevena in the region of Western Macedonia in Greece, complemented with similar data from a neighbouring variety, spoken in Kozani (also in the area of Western Macedonia in Greece).

Our data were collected using structured questionnaires. Individuals were asked to judge the grammaticality of the various syntactic constructions which involve adnominal genitives, with or without a prepositional realization. In cases of gradience, the subjects were asked to rate grammaticality on a scale from 0–6 ('0' for a perfectly bad and '6' a perfectly good sentence~~ respectively~~). Five individuals (males and females) aged from twenty-two to thirty-four were interviewed. All of them were born and raised in Grevena to at least one parent born and raised also in Grevena (or a village in the prefecture). These individuals are ~~bilingual/~~bilectal speakers of SMG and GG. As our questions often included scenaria that appeared to be quite challenging when presented to older people, i.e. non-attrited monolingual GG speakers, our data were compared to written sources documenting utterances and oral texts by monolingual elderly speakers of the Kozani dialect (from Christodoulou 2003). Comparison with data from such texts by the Kozani dialect speakers show a complete match with our findings.

The first studies that mention the recession of the inflectional genitive in Northern Greek are Kretschmer (1905) and Papadopoulos (1926),[5] while the extensive use of *apo*-PPs for true possessors (17a, 18a), which cannot be mistaken for sources, agents, or partititives, instead of inflectional genitives (17b, 18b), was already noted by Tsopanakis (1963) in relation to the dialect of Kozani and Western Macedonia in general.

(17) a. tu samar ap tu γumar (Kozani)
the.NEUT.SG.NOM/ACC saddle.NOM/ACC from the.NEUT.SG.ACC donkey.ACC[6]

b. *tu samar t γumarju
the.NEUT.SG.NOM/ACC saddle.NOM/ACC the.NEUT.SG.GEN donkey.GEN
'the saddle of the donkey'

(18) a. ta ruxa ap ta piðja (Kozani)
the.NEUT.PL.NOM/ACC clothes.NOM/ACC from the.NEUT.PL.ACC kids.ACC

[5] Thanks to an anonymous reviewer for pointing this out to us.

[6] Prepositions such as *apo* invariably take accusative complements. Here we gloss nouns, determiners, and adjectives mentioning all the cases the form would be compatible with, as the issue of case syncretism will become relevant in later sections.

OUP UNCORRECTED PROOF – FIRSTPROOFS, Wed. Feb 14 2024, INTEGRA

b. *ta ruxa ton piðjon
the.NEUT.PL.NOM/ACC clothes.NOM/ACC the.GEN kids.GEN
'the kids' clothes'
(from Tsopanakis 1963: 298, via Ntinas 2012)

11.3.1 Range of meanings and uses, iterability, and Cinque's (1980) hierarchy

As already said, what sets GG apart from SMG and other varieties is the fact that *apo*-PPs can support the full range of relations typically expressed through inflectional genitives. Thus, the PP in (19) can express a source meaning, like in SMG, but also any of the following meanings typically associated with the genitive: 'possession-control' (in the sense of Barker 2008), 'author', and any pragmatically recoverable relation between the mayor and the book (e.g. it could denote the book the mayor talked about, the book related to his term as a mayor etc.). In picture nominals like (20), the *apo*-PP can be the theme argument, i.e. the individual depicted in the picture, in addition to 'author' and 'possessor' readings.

(19) tu vivliu ap tun
the.NEUT.SG.NOM/ACC book.NOM/ACC from the.M.SG.ACC
ðimarxu (GG)
mayor.ACC
'the mayor's book'

(20) fotoγrafies ap ts peθamen' ts
pictures from the.M.PL.ACC dead.NOM/ACC the.M.PL.ACC
ðimarxei (GG)
mayors.NOM/ACC
'pictures of the dead mayors' (also: pictures shot by/owned by/coming from the dead mayors'

In our attempt to check whether prepositional genitives in GG can indeed express pragmatically recoverable relationships beyond ownership and 'part-of' relations (like Saxon *'s* genitives and unlike prepositional *of*-genitives in English), we presented our informants with a context specifically probing the pragmatically recoverable relationship 'being assigned x/being responsible for x' and asked them whether (21a–c) sounded grammatical and appropriate. All three sentences in (21) were deemed well-formed and felicitous on the intended reading. The responses to these data also suggest that the prepositional realization is compatible with either definite, quantificational, or (bare) indefinite nominals.

OUP UNCORRECTED PROOF – FIRSTPROOFS, Wed. Feb 14 2024, INTEGRA

(21) Context: 'Back in 1995, at the time of the big earthquake, we had these people from the prefecture that were checking the houses and classified them according to the level of damage incurred. Now, imagine that I examine all the houses assigned to each employee myself and I utter the following sentences'.

a. Ta spitia ap tun ipalilu taði itan
The houses from the.M.SG.ACC employee.ACC X were
indaks. (GG)
ok
'Employee X's houses were fine'

b. Ta spitia ap mirikus/miriki ipalil
The houses from some.ACC/some.NOM/ACC employees.NOM/ACC
itan indaks.
were ok
'Some employees' houses were ok.'

c. Spitia ap ipalil vreθkan akatalila.
Houses from employees.NOM/ACC were-found unsuitable
'Some employees' houses were found/deemed unsuitable.'

Another crucial property that sets PP genitives apart from SMG inflectional genitives is that PP genitives are iterable and, when iterated, ambiguous with respect to theta assignment, as opposed to inflectional genitives, which cannot be iterated in similar event nominalizations (cf. (4)).

(22) i pirighrafi ap tun ðimarxu ap
the.F.SG.NOM description from the.M.SG.ACC mayor.ACC from
ta piðja (GG)
the.N.PL.ACC kids
'the description of the mayor by the kids/the description of the kids by the mayor'

In other words, Cinque's (1980) thematic hierarchy does not affect the relative ordering of two co-occurring PP genitives. However, this does not mean that the related restrictions are suspended altogether in GG. When a PP genitive co-occurs with a (high) genitive clitic, the theta-role of the PP genitive cannot be higher with respect to Cinque's hierarchy than the theta-role of the clitic. This can be seen clearly in picture nominals, in which *apo*-PPs cannot be interpreted as external arguments in the presence of a genitive clitic (23). The PP can only correspond to the depicted individual, if the clitic corresponds to the author. If the clitic is to

OUP UNCORRECTED PROOF – FIRSTPROOFS, Wed. Feb 14 2024, INTEGRA

be interpreted as denoting the possessor, then the PP can correspond to either the author or the depicted individual.[7]

(23) i {proti/θki} m fotoγrafia ap tun ðimarxu (GG)
the first/own my.CLITIC picture from the mayor

'my first/own picture of the mayor'/'*The mayor's first/own picture of me'

The prepositional realization is only resisted when the adnominal argument is a (personal or demonstrative) pronoun (24) or a proper name (25). In such cases the inflectional realization is still favoured. In relation to (24), in particular, it is worth noting that in SMG the strong form of the personal pronoun is generally dispreferred for adnominal arguments and the clitic is the default option, unless an emphatic or contrastive interpretation is intended. Notably, whether emphatic or not, the genitive of the personal pronoun cannot be realized as a PP in GG. With all other types of nominals, however, the inflectional genitive is in fact disfavoured, at least in a purely dialectal context/register (cf. (17b–18b)).

(24) tu spiti m / *apo mena (GG)
the house my.CLITIC / from me.ACC

(25) Kaiki tu spit {t Jan} /{*/%ap
Was-burnt the house the.GEN/ACC John.GEN/ACC/ from
t Jan}.
the.GEN/ACC John.GEN/ACC

'John's house was burnt down'

In idiolects or registers with minimal or no interference of the standard variety, the only inflectional genitives that are grammatical are those of pronominal clitics and proper names, including sometimes nouns with a proper name-like function, e.g. kinship terms, in some varieties (*ts manas m* 'the.GEN mother.GEN my.CL', *t patera m* 'the.GEN father.GEN my.CL').[8] All other inflectional genitives are arguably not part of the dialectal grammar. When such inflectional genitives and prepositional genitives co-occur, a strict thematic hierarchy effect arises, with the inflectional

[7] The source interpretation of the *apo*-PP is also independently available, like in all Greek varieties. As such it does not function as a genitive and the thematic hierarchy does not apply, i.e. the clitic can (marginally) denote the depicted individual if the PP is to be interpreted as the source from which the picture came/was taken.

[8] Ntagkas 2016 observes that in one variety (Ventzia, Grevena), inflected forms of the singular genitive are also attested for few bisyllabic animal names (*t kuku* 'the.GEN cuckoo.GEN'), although he makes it clear that in that variety too the prepositional construction has prevailed, for all sorts of construction in which the genitive was used, e.g. *bábu apu kató xrónja* 'granny from hundred years.ACC, cf. *jaja ekato xronon* 'granny hundred years.GEN' in SMG). Thanks to an anonymous reviewer for bringing this work to our attention.

OUP UNCORRECTED PROOF – FIRSTPROOFS, Wed. Feb 14 2024, INTEGRA

genitive necessarily corresponding to the higher argument and the PP genitive to the lower one (26–27).

(26) i futuγrafia t Jan ap ta
the picture the.GEN/ACC John.GEN/ACC from the.NOM.ACC
piðja (GG)
kids.NOM.ACC
'John's picture of the kids'/'*the picture of John that the kids own'

(27) i piriγrafi t Jan ap ta
the description the.GEN/ACC John.GEN/ACC from the.NOM.ACC
piðja (GG)
kids.NOM.ACC
'John's description of the kids'/
'the description ~~of the kids by John~~' (if the *apo*-PP is interpreted as a by-phrase)

The unmarked placement of inflectional genitives of proper names is postnominal, before any PPs, including prepositional genitives (28).

(28) *i piriγrafi ap ta piðja t
the description from the.NOM.ACC kids.NOM.ACC the.GEN/ACC
Jan (GG)
John.GEN/ACC
'(intended) John's description of the kids'

Given these, we can safely make two assumptions. First, that inflectional genitives of proper names occupy the same position as inflectional genitives (of all sorts of nominal) in SMG. Second, the position for PP genitives is occupied by arguments that have not been attracted by a functional projection that licenses genitive Case, either because (a) the arguments that end up being realized as PPs are first merged lower than all the arguments attracted by genitive-licensing functional projections, or (b) because no genitive-licensing heads are projected. We will come back to the characteristics and the status of the position of genitive PPs very shortly.

An interesting argument for the assumption that non-proper-name inflectional genitives are not part of the GG system comes from the behaviour of bidialectal speakers, who are the majority of contemporary dialectal linguistic communities. Such speakers do accept and produce lexical inflectional genitives, but when they do so *apo*-PPs can no longer function as genitives. In other words, the two grammars may co-exist in a single speaker but cannot be mixed and only one of the two is activated. Thus, in (29) the lexical inflectional genitive cannot be interpreted as the higher of the two arguments; instead, the PP genitive is necessarily interpreted as a *by*-phrase.

OUP UNCORRECTED PROOF – FIRSTPROOFS, Wed. Feb 14 2024, INTEGRA

(29) i piriγrafi t(u) ðaskalu ap ta
the description the.GEN/ACC teacher.GEN from the.NOM.ACC
piðja (GG)
kids.NOM.ACC
'the description of the teacher by the kids'/*'the teacher's description of the kids'

11.3.2 PP genitives have the distribution of reduced relatives, not inflectional genitives

We now turn to the question of the syntactic representation of PP genitives. We will show that they display neither the syntactic behaviour of inflectional genitive DPs nor that of run-of-the-mill PPs. Instead, they are to be paralleled to reduced relatives, which also best describes the syntactic behaviour of *de-/di*-genitives in Romance. We have already contrasted GG prepositional genitives to inflectional genitives and identified two major features that differentiate them from each other: PP genitives can be iterated and a pair of PP genitives that results from such iteration is immune to any hierarchy effects (see (22)). On the other hand, they are not just like any other PP. Unlike other PPs (30), prepositional genitives can be fronted and (A'-) extracted (31). This actually brings them closer to inflectional genitives, which can also be fronted and extracted (32).

(30) *Ja pjon vaslja tu vivliu ðjavasis? (GG)
For which king the book read.2SG
'For which x, x a king, did you read a book about x?'

(31) a. [$_{DP}$ [Apu pjon]$_i$ u ksaefus t_i] efiji xte? (GG)
From whom.ACC the cousin left.3SG yesterday

b. [Apu pjon]$_i$ efiji [$_{DP}$ t_i' u ksaefus t_i] xte? (GG)
From whom.ACC left.3SG the cousin yesterday
'Whose cousin left yesterday?'

(32) a. [$_{DP}$ Pjanu$_i$ o ksaðerfos t_i] efije xtes? (SMG)
Whose the cousin left.3SG yesterday

b. Pjanu$_i$ efije [$_{DP}$ t_i' o ksaðerfos t_i] xtes? (SMG)
Whose left.3SG the cousin yesterday
'Whose cousin left yesterday?'

Nevertheless, unlike inflectional genitives, which are allowed to be fronted or sub-extracted in different kinds of nominal, including bare article-less nominals in predicate position (33), PP genitives cannot reach the edge of the phrase (and

OUP UNCORRECTED PROOF – FIRSTPROOFS, Wed. Feb 14 2024, INTEGRA

move through it), unless the nominal is clearly headed by a definite determiner (34 vs. 32).

(33) a. [$_{NumP}$ Pjanu$_i$ ksaðerfos t$_i$] ine o Kostas? (SMG)
Whose cousin is the Kostas

b. Pjanu$_i$ ine [t$_i$' ksaðerfos t$_i$] o Kostas?
Whose is cousin the Kostas

'Whose cousin is Kostas?'

(34) *Apu pjon (ksaefus) ini (ksaefus) u Kostas? (GG)
From whom.ACC cousin is cousin the Kostas

'Whose cousin is Kostas?'

Thus, in fact, the distribution of PP genitives mostly overlaps with the distribution of definite article-adjective sequences in so-called 'polydefinite' DPs (see Alexiadou and Wilder 1998, Alexiadou 2014, among many others), as far as their canonical and non-canonical placement is concerned (see also Mathieu and Sitaridou 2005 on such extractions). This is evident if one compares (31) and (34) to (35) and (36) respectively. Definite article-adjective sequences in polydefinite DPs typically involve restrictive APs or in general APs that have properties associated with so-called *indirect modification* (see Cinque 2010 about several classes of postnominal APs in Romance). We will follow the assumption (common to all aforementioned authors' works) that all such constituents are to be treated uniformly, and in particular as reduced relative clauses (cf. especially Alexiadou and Wilder 1998 and much subsequent work).

(35) (?To MEγALO) pulisa [DP (to meγalo) to spiti tis
The big sold.1SG the big the house the.GEN

Anas (to meγalo)] (SMG)
Ann.GEN the big

'I sold Ann's big/BIG house.'

(36) a. (*Meγalos) ine [$_{NumP}$ piitis tis Romjosinis
Great is poet the.GEN Greek-nation.GEN

(meγalos)] (SMG)
great

'He is a great poet of the Greek nation.'

b. (*Meγalo) pulisa [$_{DP}$ kapjo/ena spiti tis Anas
Big sold.1sg some/one house the.GEN Ann.GEN

(meγalo)] (SMG)
big

'I sold a certain/one big/BIG house of Ann's.'

OUP UNCORRECTED PROOF – FIRSTPROOFS, Wed. Feb 14 2024, INTEGRA

For reasons that we will not examine here, inflectional genitives are free to move within and from all types of nominals, both (definite) DPs and nominals of smaller size, serving as predicates. This is a property that holds of genitives in all languages where the genitive is marked on the inflection of the noun (Greek, Latin, Slavic, etc.). Presumably the inflectional features themselves are to be correlated with this flexibility (see Reeve 2019 on the relation between Agree and extraction in the extended projection of N).

Then there is a contrast between definite DPs and all other types of nominals with respect to the A'-movement possibilities of PP genitives and reduced relatives/indirect modifiers. There is a crucial difference between definite DPs, i.e. nominals headed by a definite determiner, and bare nominals, arguably projections smaller than DP (perhaps NumPs, see Alexopoulou, Folli, and Tsoulas 2013), such as the ones showing up in predicate position. In the former, there is an empty non-thematic, non-Case position at the edge of the phrase, namely the specifier of DP, while in the latter there is no such position. The specifier of DP is a position that can either host fronted phrases or provide an escape hatch for phrases sub-extracted from the DP (see Horrocks and Stavrou 1987). In the light of our discussion of inflectional genitives here, this escape hatch is specifically needed for phrases not inflectionally marked by the head noun. The specifier of lower projections, such as NumP, is not an A'-position and cannot function as an escape hatch (34a, 36a). Such an escape hatch is also unavailable in DPs with determiners other than the definite article. We have to assume that all these determiners are actually phrasal, as they can be further modified and coordinated: all cardinal numerals, for instance, can be followed by *ke miso* ('and a half'), which would be impossible if the determiner occupied a head position. Therefore, all determiners except the definite article occupy Spec-DP themselves and do not make this position available as an escape hatch, see (34b) and (37) from GG.

(37) *Apu pjon$_i$ iðis [kapjun/enan ksaefu t$_i$]? (GG)
From who.ACC saw.2SG some/one.ACC cousin.ACC

'For which x, x a guy, did you see some/one cousin of x?'

Table 11.1 summarizes the properties of PP genitives, inflectional genitives, non-genitive PPs, and reduced relatives/indirect modifier APs with respect to fronting and extraction from definite DPs and other nominals, iterability, and their unmarked position.

From a cross-linguistic perspective, it is worth contrasting the Northern Greek prepositional genitive to prepositional genitives of other languages, such as English *of*-genitives and Romance *de-/di*-genitives. It turns out that English *of*-genitives are not easily iterable (38a) and cannot easily be made compatible with all possible thematic interpretations in all contexts (38b). Notably, as opposed to Saxon *'s* genitives, which can also denote non-possessor pragmatically recoverable relationships, as already mentioned, such a relationship cannot

OUP UNCORRECTED PROOF – FIRSTPROOFS, Wed. Feb 14 2024, INTEGRA

Table 11.1 PP genitives contrasted to similar constituents

	PP genitives	**Inflectional genitives**	**Reduced relatives/ Indirect modifiers**	**Other PPs**
Fronting/extraction from within DPs	**Yes**	Yes	**Yes**	No
Fronting/extraction from within nPs	**No**	Yes	**No**	No
Iterability	**Yes**	No	**Yes**	Yes
Unmarked rightmost position	**Yes**	Yes	**Yes**	Yes

be expressed by *of*-genitives (see Barker 2008). In contrast, *de-/di*-genitives in languages such as Italian, French, and Spanish, are as freely iterable (39a) as *apo*-genitives (cf. Androutsopoulou and Echevarría 2003) and compatible with loose pragmatic relationships between the head noun and the adnominal genitive (39b). Also, when iterated (39a), Romance-type prepositional genitives exhibit no strict ordering effects, only slight preferences (cf. Sportiche 1992 for French).

(38) a. *the description of John of Mary
b. the picture of John

'the picture depicting John/*shot by John/*owned by John/that John had to take care of'

(39) a. (?)la descrizione del sindaco dei ragazzi
the description of-the.M.SG mayor of-the.M.PL kids

'the mayor's description of the kids'/¿the children's description of the mayor'

b. il ritratto di Giovanni
the portrait of Giovanni

'the portrait depicting Giovvani/painted by Giovanni/owned by Giovanni/that was assigned to Giovanni'

It is also worth noting that indirect modifier APs in Italian, i.e. postnominal APs with a restrictive interpretation, can either precede or follow a *di*-PP (40). When preceding the *di*-PP, postnominal APs are ambiguous between a restrictive and a non-restrictive interpretation. They can thus be analysed as either the result of n(P)-movement over the AP, which is compatible with a non-restrictive interpretation, or as true, necessarily restrictive, reduced relatives (Cinque 2010). When following the *di*-PP, postnominal APs have to be stressed and are only compatible with a restrictive interpretation, i.e. they can only be reduced relatives. Therefore, the base position of direct modifier PPs can never be below the *di*-PP and, more

OUP UNCORRECTED PROOF – FIRSTPROOFS, Wed. Feb 14 2024, INTEGRA

importantly, reduced relatives and *di*-PPs are not strictly ordered with respect to one another. Given these, it is reasonable to assume that *di*-PPs and indirect modifier APs have the same or similar structural representation and origin, perhaps with both of them originating as reduced relatives.

(40) a. Gli industriosi greci di Megara di sicuro saranno
The industrious Greeks of Megara certainly will-be
premiati. (Italian)
awarded-a-prize

b. I greci industriosi di Megara di sicuro saranno
The Greeks industrious of Megara certainly will-be
premiati.
awarded-a-prize

c. I greci di Megara industriósi di sicuro saranno
The Greeks of Megara industrious certainly will-be
premiati.
awarded-a-prize

'The industrious Greeks of Megara will certainly receive a prize.'
(from Cinque 2010: 79)

Cinque (2010) analyses reduced relatives as base-generated higher than the extended projection of *n* that includes direct modifier APs. Their phrase-final placement in Italian (as well as Greek)[9] is due to massive movement of the extended projection over the position of reduced relatives, an insight we also adopt in our representation in (41) below.

Overall, in the light of the overall behaviour of adnominal arguments in GG and in particular the mentioned parallels between prepositional genitives in GG (or, in general, NG) and Romance prepositional genitives, we can conclude that the structure of the DP in GG (41) contains the following positions for adnominal genitive arguments: (a) a slot for prepositional genitives, structurally similar to or the same as the position of *de-/di*-genitives, in the domain reserved for reduced relatives (RR in the representation below), (b) F^0_{CLITIC} for genitive pronominal clitics, just like in SMG, but also (c) a residual F^0_{GEN} just for proper names and postnominal genitive pronominal clitics. This state of affairs is in fact

[9] The phrase-initial placement of 'definite article-adjective' sequences in Greek, which is possible alongside phrase-final placement, is the result of further movement of the reduced relative over D (see Alexiadou 2014 and references therein), after the extended projection's massive movement à la Cinque. This further movement is what the Romance languages lack, so the fact that *di*-PPs do not front as easily and do not get extracted at all does not come as a surprise.

OUP UNCORRECTED PROOF – FIRSTPROOFS, Wed. Feb 14 2024, INTEGRA

reminiscent of Romance varieties which have developed a full-fledged system of prepositional genitives, after the decline and loss of genitive/oblique case-marked nouns, but still retain a residual postnominal genitive construction, just for a limited number of noun classes (see Silvestri 2013 for such a dialect, spoken in Verbicaro, Northern Calabria).

(41) $[_{DP}$ D $[$ $[_{F1P}$ (Adj) $[$ $F^0{}_{CLITIC}$ (Adj) n $[$ $DP_{GEN.proper\ name/pronoun}$ $F^0{}_{GEN}$ $[\ldots\sqrt{\ldots}]]]]$ $[_{F2P}$ PP-gen/RR F_2 t_{F1P} $]]]$[10]

11.4 The morphological correlates of the rise of prepositional genitives

The dialects of Greek provide us with the ideal experimental setting whereby two sets of varieties may differ in their syntax to such a minimal degree that any other differences that cluster together are unlikely to be unrelated. The Northern Greek dialects that exhibit extensive use of prepositional genitives are also characterized by relatively few other differences from SMG. The most notable differences are the northern dative-accusative syncretism (as opposed to the dative-genitive syncretism of SMG) and certain restructurings in the declensional paradigm of nouns and adjectives. The dative-accusative syncretism may have played some role, as also suggested by Mertyris (2014), in reducing the frequency of genitive forms in the input of acquirers of these dialects, in comparison to SMG and in general central/southern varieties. Nevertheless, other changes concerning the inflectional suffixes of the nominal domain may have had some broader and more relevant consequences. In particular we will draw a correlation between the retention/loss of inflectional genitives and the presence/absence of case-number fusion and case-conditioned stem allomorphy.

In SMG, the inflectional suffixes of nouns are commonly analysed as portmanteaus in which [Case] and [Number] features are fused. Ralli (1994) distinguishes between eight inflectional classes (Table 11.2). Among them, we can further discern two broad categories of inflectional classes: (A) classes that comprise nouns whose stem does not vary depending on the suffix or may vary for purely phonological reasons; (B) classes with number- and case-conditioned stem allomorphy.

In fact, all four declensional classes that contain neuter nouns (Ralli's Classes V–VIII) belong to category A. In addition, all neuters exhibit a nominative-accusative syncretism in the singular and the plural. In other words, the existence

[10] This representation involves massive movement of the constituent (FP) that contains the noun and projections up to $F^0{}_{CLITIC}$ and the highest adjectives. This constituent must have originated below the domain of reduced relatives, following Cinque (2010).

OUP UNCORRECTED PROOF – FIRSTPROOFS, Wed. Feb 14 2024, INTEGRA

Table 11.2 Ralli's (1994) eight inflectional classes (adapted from Alexiadou and Müller 2008)

	I: masc *kip* 'garden'	**I: fem** *psif* 'vote'	**II: masc** *maxit(i)* 'fighter'	**III: fem** *avl(i)* 'yard'	**IV: fem** *pol(i/e)* 'city'	**V: neut** *vun* 'mount'	**VI: neut** *krat* 'state'	**VII: neut** *spiti* 'house'	**VIII: neut** *soma(t)* 'body'
nom/sg	os	os	s	Ø	Ø	o	os	Ø	Ø
acc/sg	o(n)	o(n)	Ø	Ø	Ø	o	os	Ø	Ø
gen/sg	u	u	Ø	s	s	u	us	u	os
nom/pl	i	i	es	es	is	a	i	a	a
acc/pl	us	us	es	es	is	a	i	a	a
gen/pl	on	on	on	on	on	on	on	on	on

of genitive forms is the only reason to consider their suffixes as a fusion of Number and Case features. If it were not for the genitive, we would say that each class has one suffix for the singular and one for the plural, i.e. suffixes realizing Number only. Therefore, for all neuters the loss of the genitive would give rise to classes with no case-number fusion and no case-conditioned allomorphies.

Category A also includes Class I nouns. In this class, suffixes clearly exhibit case-number fusion, as there is no unique segment or sequence of segments that can be associated with number only or with case only, even putting the genitive aside. Category B classes exhibit allomorphic stems, whose allomorphy does not depend on phonological properties of either the stem or the suffix. As shown in the paradigms below (42–44), in Classes II and III there is a distinction between singular allomorphs and plural allomorphs, while in Class IV there is a singular allomorph, and two plural allomorphs, with further sensitivity to Case.

(42) **Class II**

Nom/sg	maxiti	-	s
Acc/sg	maxiti	-	Ø
Gen/sg	maxiti	-	Ø
Nom/pl	maxit	-	es
Acc/pl	maxit	-	es
Gen/pl	maxit	-	on

(43) **Class III**

Nom/sg	avli	-	Ø
Acc/sg	avli	-	Ø
Gen/sg	avli	-	s
Nom/pl	avl	-	es
Acc/pl	avl	-	es
Gen/pl	avl	-	on

OUP UNCORRECTED PROOF – FIRSTPROOFS, Wed. Feb 14 2024, INTEGRA

(44) **Class IV**

Nom/sg	poli	-	Ø
Acc/sg	poli	-	Ø
Gen/sg	poli	-	s
Nom/pl	pol	-	is
Acc/pl	pol	-	is
Gen/pl	pole	-	on

Interestingly, if we were to delete all genitive cells, a different segmentation of some endings would be possible (45). Specifically, it would be possible to consider *-s* as a form realizing plural Number in all three classes. The *-e-* of *-es* endings (Class II and III) could now be considered a theme vowel, alongside the *-i-* of *-is* in the plurals of Class IV. The form of the theme vowel would only be sensitive to Number (and, of course, inflectional class). Furthermore, the only allomorph which is exclusively sensitive to Case, namely the plural genitive allomorph of Class IV (*pole-*) would disappear and then all allomorphy would be exclusively conditioned by Number.[11]

(45)	Root	Theme vowel	Number	Case	
Class II	maxit	i	sg. Ø	nom.	-s
				acc.	Ø
		e	pl. s	nom.	-s/Ø[12]
				acc.	Ø
	iro	a	sg. Ø	nom.	-s
				acc.	Ø
		e	pl. s	nom.	-s/Ø
				acc.	Ø
Class III	avl	i	sg. Ø	nom.	Ø
				acc.	Ø
		e	pl. s	nom.	Ø
				acc.	Ø

[11] Alternatively, one might conclude that these theme vowels actually now realize Number themselves. However, this would complicate the role of -s. There is no reason to think that it encodes Case in any meaningful way, except in the singular nominative of Class II.

[12] If we assume that the nominative suffix is -s for both numbers, we predict that the underlying representation of Class II plural nominatives is *maxit-e-s-s*. The simplification leading to /maxites/ is not uncommon in SMG, cf. /kateðafiz - si/ ➔ /kateðafis - si/ ➔ /kateðafisi/, as SMG has no geminate consonants. Therefore, postulating a *-s* nominative suffix for Class II, common to both singular and plural, should not be a problem for acquirers/native speakers. Otherwise, we can posit contextual allomorphy and assume that the vocabulary item realizing Case is sensitive to Number, so that the vocabulary item selected in the context of Singular Number (and Class II) is */-s/*, while Ø is the vocabulary item inserted in the context of Plural Number or as the result of an Elsewhere rule. This analysis has the advantage of making Class II Case suffixes identical to the ones of Class I (cf. (48)).

OUP UNCORRECTED PROOF – FIRSTPROOFS, Wed. Feb 14 2024, INTEGRA

Class IV	pol	i	sg. Ø	nom.	Ø
				acc.	Ø
		i	pl. s	nom.	Ø
				acc.	Ø

So far, we have seen how the loss of genitive inflection from the input would lead the acquirers to a state of affairs whereby they need not postulate any number-case fusion or any case-sensitive allomorphy, for all inflectional classes except one. The one recalcitrant class in this respect is Class I (46). Even if the genitive is no longer part of the paradigm, the plural suffixes are still such that no further segmentation is possible, i.e. case and number appear to be fused.

(46) Class I (SMG)

Nom/sg	kip-	os
Acc/sg	kip-	o
Nom/pl	kip-	i
Acc/pl	kip-	us

From this perspective, the restructuring of exactly this class in many Greek dialects, including GG, Kozani Greek, and indeed most Northern Greek dialects, is of great importance. In these dialects, the plural nominative of Class I nouns now coincides with the plural accusative (47).

(47)

Class I (NG)	Definite article	stem	ending
Nom/sg	o/u	kip-	os/us[13]
Acc/sg	ton/tun	kip-	o/u
Nom/pl	i	kip-	(i)
Acc/pl	t(u)s	kip-	(i)

Regardless of this restructuring, this class can in fact be paralleled to Class II and speakers may postulate the same singular suffixes for both. Unlike Ralli (1994) and analyses based on her classification, Revithiadou, Spyropoulos, and Markopoulos (2017) propose that the singular endings of Class II can in fact be split, with the *-o-* being associated with Number only. The rest, namely *-s*, *-Ø*, *-i*, *-us* are number-case portmanteaus. We will adopt this analysis of *-o-* for SMG and NG alike. In addition to this, in NG, in the plural too there is a vowel common to both nominative and accusative, namely *-i*. Therefore, there is no reason to think of the plural endings as fusional any longer (48).

[13] Due to a cluster of phonological phenomena known as 'northern vocalism' (on which see the discussion after (54)), what would be realized as unstressed [o] in SMG is normally realized as [u] in NG. Our informants, being younger bilectal speakers, are not completely consistent with respect to this, with [o] also being a possible (though less frequent) realization. We will not take a position here as to the underlying phonological representation, though see some speculations under (54), where the issue becomes relevant.

OUP UNCORRECTED PROOF – FIRSTPROOFS, Wed. Feb 14 2024, INTEGRA

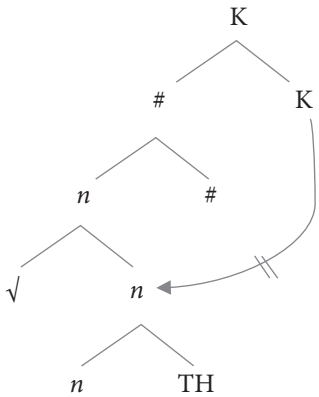
(48)

	Root	Theme vowel	Number	Case	
Class II	kip		sg. o	nom.	-s
				acc.	-Ø
			pl. i	nom.	-Ø[14]
				acc.	-Ø

The emerging picture is compatible with a structural representation whereby Number and Case project separately both in the syntax and at PF. The retention of inflectional genitives prevents them from doing so for two reasons: (a) firstly because singular genitive endings in Class I nouns and all plural genitive endings are necessarily fusional, thus indicating that the two features cannot project separate nodes, and (b) secondly because there is a stem allomorph which only shows up in the plural genitive of Class IV (e.g. *pole-* 'city'), i.e. it is sensitive to case as well as number, while also the theme vowel *-o-* of Class I nouns disappears not only in the plural but also in the genitive of the singular, with the latter being exclusively case-conditioned.

The facts in (b) in particular entail Number-Case fusion for the following reason: in order for a feature X to condition allomorphy in the realization of a lower node Y, X and Y must be local enough. In our case (cf. 45), allomorphic X is a theme vowel TH associated with/attaching to *n*, i.e. a cyclic node. Following Embick (2010: 68), a cyclic node X cannot have its allomorphy determined by a noncyclic node Y unless 'they are concatenated when Vocabulary Insertion occurs'. If Number (Num) and Case (K) projected separately, each with its own overt exponent (49), then Case should not be able to determine the allomorphy in *n*. Only when they are fused is such sensitivity to Case allowed (50). In both (49) and (50) we follow Revithiadou et al's (2017) treatment of theme (TH) vowels as adjoining to *n*.

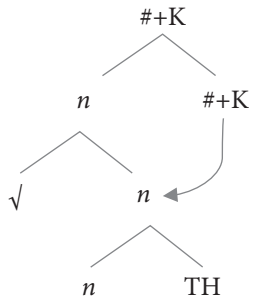
(49)



[14] See fn. 12 regarding the vocabulary insertion rule for the nominative Case morpheme in the context of plural Number. If this rule is common to both Class I and Class II nouns, then both classes have identical sets of case suffixes in NG and the VI rule for /-s/ would concern both Class I and Class II.

OUP UNCORRECTED PROOF – FIRSTPROOFS, Wed. Feb 14 2024, INTEGRA

(50)



Input that contains Number-Case portmanteaus as well as Case-conditioned stem allomorphy necessarily makes acquirers postulate a node in which Case and Number are fused. On the other hand, there appears to be a bias/economy constraint guiding language acquisition and language change which favours representations with as few features per head/node as possible (51).

(51) A structural representation R for a substring of input text S is simpler than an alternative representation R′ iff R contains fewer formal feature syncretisms[15] than R′.
(Roberts and Roussou 2003: 201)

After the restructuring of Class I plurals in GG and other NG varieties, it was easier for acquirers to succumb to the pressure for fewer 'feature syncretisms'. Prior to their generalized use for all nouns, PP genitives must have started their life as suppletive forms for certain nouns, like in other dialects. However, given the lack of any other forms beyond the genitive that would speak in favour of Number-Case fusion, acquirers were now freer to consider *apo*-PPs as the main exponent of the genitive, exclude from all paradigms any remaining forms with case-conditioned stem allomorphy and fusional suffixes, i.e. all genitive forms, and finally postulate the representation in (49) for all inflectional classes. Therefore, the following one-way implication emerges from this reasoning: the loss of the inflectional genitive and its wholesale replacement by adpositional genitives is possible only if the rest of the paradigm contains no cues for Number-Case fusion. This is confirmed by the fact that within Greek, all dialects with predominantly prepositional genitives are dialects where this restructuring of Class I has taken place, but not all dialects with nominative-accusative syncretism in Class I have generalized the use of prepositional genitives yet (see e.g. Lesbian or Italiot Greek).

The correlation drawn here seems to hold on a cross-linguistic/typological level as well. In what follows, we present an argument in favour of this correlation,

[15] In Roberts and Roussou (2003) 'feature syncretism' is understood as the coexistence of more than one formal feature on the same head.

OUP UNCORRECTED PROOF – FIRSTPROOFS, Wed. Feb 14 2024, INTEGRA

which can be built using the tools of the World Atlas of Language Structures (WALS, Dryer and Haspelmath 2013). It is reasonable to assume that languages in which all non-core cases, i.e. cases such as genitive, dative, instrumental etc., are realized adpositionally cannot have more than two cases marked inflectionally (typically, nominative-accusative, absolutive-ergative, or direct-oblique). Thus, languages that can be identified as having adpositional case are languages that WALS characterizes either as languages with 'no morphological case-marking' or as languages with '2 cases', with respect to the feature 'Number of Cases'. The other relevant WALS features is 'Exponence of Selected Inflection Formatives', which mainly provides information on the forms realizing case. There are 129 languages in the database for which a value is provided with respect to both the feature the feature 'Exponence of Selected Inflection Formatives' and 'Number of cases'. Of them, seven languages are marked as combining 'monoexponential case' (i.e. case is never fused with any other feature) with 'no morphological marking' (i.e. cases, if there are any, can only be adpositional). Another eight languages combine 'monoexponential case' with '2 cases' (i.e. all non-core cases are adpositional, if there are any such cases). Among languages with 'Case + number syncretism', no languages have the value 'no morphological marking' (i.e. none express their case exclusively through adpositions) and there is only one language marked as having '2 cases'. This language is Yaqui, which turns out to have two morphological cases, one for possessors and one for non-possessors, i.e. what looks like a genitive is in fact inflectional. Also, languages that fuse Case with other features, such as referentiality or Tense-Aspect-Mood, tend to have a big number of inflectional cases (and therefore no adpositional cases). In fact there are no such languages with fewer than ten inflectional cases. This typological picture confirms the one-way implication we derived in the previous paragraph, repeated here as a tentative typological universal (52).

(52) **The adpositional case generalization**: A language can have adpositional case-markers for a case (for a [+N] category X) only if there are no fusional case-number exponents for any other cases (for X).

This correlation also points to a diachronic scenario/pathway regarding the other major cue of Number-Case fusion, namely case-sensitive stem allomorphy: adpositional genitives are generalized and prevail once all case-conditioned stem allomorphy is lost or on its way out for all cases. This of course also includes the former/declining (inflectional) exponent of the genitive, if it happened to display case-conditioned stem allomorphy. Indeed, the history of French is telling in this respect. Old French had two cases (*cas sujet*/direct and *cas régime*/oblique), each of which was often associated with a distinct stem allomorph. Indeed, the rise of *de*-PPs and their becoming the exclusive exponent of the genitive historically correlates with the loss of one of the two stems for nouns with irregular declension

OUP UNCORRECTED PROOF – FIRSTPROOFS, Wed. Feb 14 2024, INTEGRA

(see Nikiforidou 1991 and references therein). Specifically, in most cases only the stem used in oblique case forms survived or, in few cases, the two stems, i.e. the nominative stem and the oblique stem, became two distinct lexemes (e.g. *pute* and *putain*, the former a coarser synonym of the latter).

In contrast, due to the relatively high number of cases and declensional classes in Slavic, the unambiguously fusional case-number suffixes are arguably too many to support a reanalysis that would favour monoexponential case. This reinforces the idea that the retention of inflectional endings for non-core cases is only possible if the rest of the nominal paradigm provides cues for nodes fusing case and number. At the same time, the loss and the replacement of non-core cases presupposes the increased compatibility of the declensional paradigm as a whole with a reanalysis of the sort previously described.

The considerations we have presented regarding the role of fusion and locality in case-conditioned stem allomorphy also explain why determiners, including pronouns, and proper names retain their inflectional genitives and still resist the prepositional realization. In determiners and pronouns, there is ample evidence that the stem suppletes for case as well as number, see for instance first-person personal pronouns (53).

(53)

1st person	Singular	Plural
Nom	eɣo	em-is
Acc/Gen	em-ena	em-as
Acc clitic	m-e	m-as
Gen clitic	m-u	m-as

2nd person	Singular	Plural
Nom	esi	es-is
Acc/Gen	es-ena	es-as
Acc clitic	s-e	s-as
Gen clitic	s-u	s-as

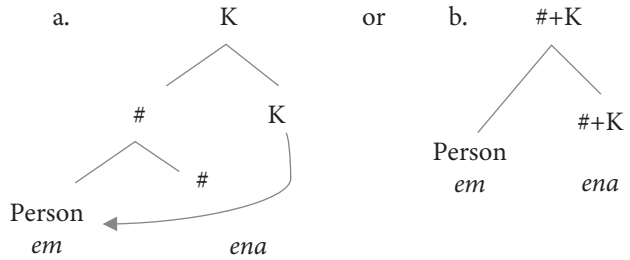
In fact, there is case-conditioned allomorphy affecting the singular accusative/genitive form. In other words, even if the genitive were to be replaced by a prepositional substitute, there would still be enough input to postulate Number-Case fusion. Given this input, there is no particular pressure or reason to reduce the inflectional realization of the genitive in particular.

Notably, in the case of personal pronouns, we do not simply have different exponents for so-called ornamental morphology such as theme vowels, but ~~what looks like~~ stem suppletion conditioned by Case, of the kind we never see in noun morphology. Following Moskal (2015b), the straightforward explanation is that ~~in D elements such as (personal) pronouns,~~ there is no *n*, which means that there is no cyclic node between Case and Person, which is essentially the node hosting what we called the stem in these Cases. Following Embick (2010: 67), a noncyclic node

OUP UNCORRECTED PROOF – FIRSTPROOFS, Wed. Feb 14 2024, INTEGRA

Z (like Person, in our case) 'may have its allomorphy conditioned by noncyclic W in … *x] Z] …] W]* provided that Z and W are concatenated when Vocabulary Insertion occurs'. Whether no VI is inserted under # (54a) or # is fused with K (54b), Person and (#+)K exponents are indeed concatenated.

(54) a. b.



The retention of Case-conditioned stem allomorphy and therefore of the inflectional genitive in proper names can receive an equally straightforward explanation. In fact, there are two possibilities. On the first scenario, we could assume that, being used to refer to non-plural individuals, proper names do not project Number at all. Therefore, K is indeed not fused/monoexponential. Given this, Case-conditioned allomorphy of theme vowels in (genitive) forms of proper names, specifically those that belong to Class I (55), does not constitute a cue for 'formal feature syncretism' and there is no pressure against it. There is also a second scenario that emerges if we capitalize on the fact that vowel raising, which is very common in NG, often renders the realization of underlying /o/ and underlying /u/ identical. This isogloss is known as 'northern vocalism' and includes deletion of unstressed /i/ and /u/ and raising of unstressed /e/ and /o/ to [i] and [u] respectively. Then, if the theme vowel of nominative and accusative forms is taken to be /u/, rather than /o/, then effectively all case-conditioned stem allomorphy vanishes.

(55)

Nom.	Nik – o – s	['nikus]
Acc.	Nik – o – Ø	['niku]
Gen.	Nik – Ø – u	['niku]

Of course, the overlapping pronunciation of the singular genitive and the singular accusative is also true of common nouns of Class I. Nevertheless, as opposed to proper names, such common nouns also have productive plurals and the plural genitive, if retained, constitutes an unambiguous cue for Number-Case fusion. This explains the considerably higher pressure against inflectional genitives of common nouns in NG compared to that of proper names.

OUP UNCORRECTED PROOF – FIRSTPROOFS, Wed. Feb 14 2024, INTEGRA

11.5 Conclusion

Despite the continuous presence of a morphologically distinct genitive case throughout the history of Greek and its varieties, recently a number of Greek varieties have developed a prepositional construction that can replace the inflectional genitive in all of its meanings and for all sorts of nouns. A detailed examination of its distribution suggests that we should not treat it as a new exponent of the same syntactic position/function and certainly not as a means of mere suppletion, i.e. as a way of filling paradigmatic gaps, but rather as a totally new construction, mostly akin to adpositional genitives in Romance. Taking a closer look at Northern Greek and contrasting it to the rest of the Greek diasystem, one can also address the so far poorly understood issue of how and when morphological case is retained or lost/replaced. We have concluded that there are certain morphological prerequisites in order for PP genitives to completely take over as an autonomous full-fledged syntactic mechanism. Specifically, for PP genitives to prevail, there should be no other potential cues for a fused representation of Case and Number (or any other feature) under the same node. This line of thought also explains the pattern of genitive replacement in Western Romance, as well as why the emergence of PP genitives was resisted by non-Northern Greek varieties, and possibly why it was resisted by all the fusional languages that still have inflectional genitive endings, in general, e.g. most of Slavic.